

Day 12 – Detection Rules & Use Case Development

Objective

The objective of this session is to develop a comprehensive understanding of how **Security Operations Center (SOC) analysts design detection rules and build use cases** to proactively identify cyber threats.

By the end of this session, learners will be able to:

- Understand the concept of detection rules in SIEM environments
- Develop structured use cases for threat detection
- Create detection logic based on real-world attack patterns
- Reduce false positives through effective rule tuning
- Map detection strategies to MITRE ATT&CK techniques
- Apply a proactive approach to threat monitoring and incident detection

1. What is a Detection Rules

A **detection rule** is a logical condition configured within a **SIEM (Security Information and Event Management)** system to identify suspicious, anomalous, or malicious activities from collected log data.

Detection rules enable **real-time threat identification** by continuously analyzing logs from multiple sources such as:

- Authentication systems
- Network devices (firewalls, IDS/IPS)
- Endpoints
- Applications
- Cloud services

➤ Role of Detection Rules in SOC

Detection rules serve as the **foundation of security monitoring** by:

- Identifying potential security incidents
- Triggering alerts for analyst investigation
- Automating initial threat detection
- Reducing manual log analysis effort

➤ Key Components of a Detection Rule

Component	Description
Data Source	Origin of logs (e.g., authentication logs, firewall logs)
Event Type	Specific activity being monitored
Condition/Logic	Rule criteria (threshold, pattern, anomaly)
Time Window	Duration for evaluation
Threshold	Limit for triggering alert
Alert Action	Notification or response triggered

2. Example Detection Rule

IF failed_login_attempts > 5
FROM same IP
WITHIN 2 minutes
THEN trigger alert: Possible Brute Force Attack

➤ Technical Breakdown

- **Data Source:** Authentication logs
- **Condition:** Multiple failed login attempts
- **Threshold:** More than 5 attempts
- **Time Constraint:** Within 2 minutes
- **Outcome:** Alert generated for SOC analyst

➤ Challenges in Detection Rules

- High number of **false positives**
- Difficulty in detecting **low-and-slow attacks**
- Lack of context in isolated alerts
- Constant need for tuning and improvement

➤ Explanation

- Monitors authentication logs
- Detects repeated failed login attempts
- Identifies potential brute force behavior
- Triggers alert for SOC investigation

◆ Detection Engineering Approach

Detection engineering is the process of **designing, testing, and improving detection logic**.

➤ Detection Lifecycle

1. Requirement Identification
2. Rule Creation
3. Testing & Validation
4. Deployment in SIEM
5. Monitoring & Feedback
6. Continuous Improvement

➤ Types of Detection Logic

- **Signature-Based Detection** → Known attack patterns
- **Anomaly-Based Detection** → Deviations from normal behavior
- **Behavioral Detection** → User/system activity analysis
- **Correlation Rules** → Combining multiple events

3. What is a Use Case?

A **use case** is a structured detection scenario that defines:

- What activity should be monitored
- How it should be detected
- What response action should be taken

Use cases provide a **complete detection strategy**, combining multiple detection rules, context, and response procedures.

➤ Importance of Use Cases in SOC

- Standardizes threat detection
- Improves investigation in SOC operations
- Aligns detection with business risks
- Supports proactive threat hunting
- Enable faster incident response
- Align detection with organizational risk

➤ Structure of a Use Case

Component	Description
Use Case Name	Name of the attack scenario
Data Source	Logs used for detection
Detection Logic	Conditions for identifying threat

Threshold	Limit or trigger value
Response	Action taken by SOC

4. Example Use Case 1: Brute Force Detection

Use Case Name: Brute Force Attack Detection

Description :

- Detect repeated failed login attempts indicating password guessing attacks.

Data Source :

- Authentication Logs

Condition :

- Multiple failed login attempts from a single IP

Detection Logic :

- More than 5 failed attempts
- Same IP address
- Within a short time window

Indicators :

- Repeated login failures
- Same username/IP combination

Response Actions :

- Generate alert
- Block suspicious IP address
- Lock user account
- Notify SOC analyst
- Investigate login patterns

Severity :

- Medium to High

4. Example Use Case 2 – Phishing Detection

Use Case Name: Phishing Email Detection

Description :

Identify phishing attempts targeting users through deceptive emails containing malicious links or attachments.

Data Sources :

- Email gateway logs
- Web proxy logs
- DNS logs

Detection Logic :

- Suspicious domains or URLs
- Email sender mismatch
- High-volume email distribution
- Known malicious indicators

Indicators of Compromise (IOCs) :

- Fake domain names
- Shortened URLs
- Attachments with macros
- Urgent or threatening language

Response Actions :

- Block domain or sender
- Quarantine email
- Remove emails from inbox
- Alert affected users
- Conduct awareness training

Severity :

- High

Detection Rule vs Use Case

Detection Rule	Use Case
Specific logic/condition	Complete detection scenario
Focuses on one pattern	Covers full attack detection
Triggers alerts	Defines monitoring + response

5. Mapping Detection to MITRE ATT&CK

Mapping detection rules to MITRE ATT&CK helps analysts understand **where the attack fits in the attack lifecycle**.

Examples

Attack Type	Tactic	Technique
Brute Force	Credential Access	Password Guessing
Phishing	Initial Access	Phishing
Malware Execution	Execution	User Execution

➤ Benefits of MITRE Mapping

- Understand attacker intent
- Improve detection coverage
- Correlate multiple alerts
- Enhance threat intelligence integration

➤ Analytical Advantage

- Provides context to alerts
- Improves incident prioritization
- Enhances threat intelligence integration

6. Best Practices for Detection Engineering

Effective detection requires continuous improvement and precision.

1. Avoid False Positives

- Avoid overly broad rules
- Fine-tune thresholds and conditions
- Use accurate thresholds
- Apply contextual filters

2. Use Clear conditions

- Define precise conditions
- Avoid ambiguous detection patterns
- Avoid overly generic rules

3. Continuously improve rules

- Update rules based on new threats
- Analyze past incidents to refine detection
- Review detection performance

4. Map detection to MITRE

- Align detection with known attacker techniques
- Improve visibility into attack stages

5. Contextual Awareness

- Consider user behavior and environment
- Avoid triggering alerts for normal activity

Assignment

1. Create One Detection Rule

Detection Rule: Suspicious Login Activity

IF login_attempts > 10
FROM same user
WITHIN 5 minutes
AND login_success = TRUE
THEN trigger alert: Possible Account Compromise

Explanation

- Detects abnormal login attempts followed by success
- Indicates potential brute force success or credential compromise

2. Write a use case for any attack

Use Case 1 : Phishing Email Detection

Data Source:

- Email logs
- Web proxy logs

Detection Logic:

- Presence of suspicious URLs
- Known malicious domains
- Emails with urgent or threatening language

Conditions:

- URL flagged in threat intelligence
- Multiple users receiving similar emails

Response Actions:

- Block malicious domain
- Quarantine email
- Alert SOC analyst
- Conduct user awareness notification

Use Case 2 : Malware Detection

Use Case Name: Malware Execution Detection

Data Source: Endpoint logs

Detection Logic:

- Execution of unknown processes
- Suspicious file behavior

Response:

- Isolate system
- Scan endpoint
- Block malicious file

3. Mapping it to MITRE ATT&CK

Activity	Tactic	Technique
Suspicious email	Initial Access	Phishing(Malware Execution)
Malicious link click	Execution	User Execution
Credential submission	Credential Access	Credential Harvesting(Credential Dumping)

Conclusion

Detection rules and use cases are essential components of **modern SOC operations**.

They enable organizations to:

- Detect threats proactively
- Automate security monitoring
- Respond quickly to incidents
- Reduce risk exposure

A well-designed detection strategy transforms SOC operations from **reactive alert handling to proactive threat detection**.

Key Takeaway

Effective cybersecurity detection is not just about creating alerts —